

Algebra

Linear to abstract, computational to structural, classical to equivariant

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Abstract

We will cover algebra from linear and computational foundations to abstract algebra, representation theory, computational algebra, and topological applications.

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1 PART I — LINEAR ALGEBRA (THEORETICAL)

1.1 1. Vector Spaces

- Fields and definitions; vector spaces and subspaces
- Span, linear independence; bases and dimension
- Sum and direct sum of subspaces
- Quotient spaces; first isomorphism theorem for vector spaces
- Why this matters: all of PDE, signal processing, and ML live in vector spaces

1.2 2. Linear Maps

- Maps as the primary object, matrices as secondary
- Null space and range; rank-nullity theorem
- Isomorphisms; space of linear maps $L(V, W)$
- Quotient constructions; exact sequences (preview)
- Applications: linear systems, PDE operators, neural network layers

1.3 3. Matrices as Representations

- Matrix of a linear map in a basis; change of basis; similarity
- Row reduction as a computational tool; rank from the map perspective
- Equivalence vs. similarity vs. congruence
- Connection to numerical linear algebra (Part II)

1.4 4. Dual Spaces

- Linear functionals; dual space V^* ; examples
- Dual basis; double dual and canonical isomorphism
- Transpose of a linear map; annihilators
- Reflexivity; connection to functional analysis and weak formulations in PDE

1.5 5. Eigentheory

- Eigenvalues and eigenvectors; characteristic polynomial
- Cayley-Hamilton theorem; consequences for matrix polynomials
- Invariant subspaces; triangularization over $\mathbb{C}$

- Generalized eigenvectors and Jordan normal form
- Minimal polynomial; relation to Jordan structure
- Applications: stability of ODEs, Google PageRank, PCA, Markov chains

1.6 6. Inner Product Spaces

- Inner products and norms; Cauchy-Schwarz
- Gram-Schmidt orthogonalization
- Orthogonal complements and projections; best approximation
- Adjoint T^* ; normal, self-adjoint, unitary, and orthogonal operators
- Applications: least squares, QR factorization, signal decomposition

1.7 7. Spectral Theory

- Spectral theorem: real symmetric (orthogonal diagonalization)
- Spectral theorem: normal operators over $\mathbb{C}$
- Positive semidefinite operators; Cholesky as spectral factorization
- Singular value decomposition: existence, uniqueness, geometry
- Applications: PCA, dimensionality reduction, pseudoinverse, graph analysis

1.8 8. Multilinear Algebra

- Bilinear forms and quadratic forms; Sylvester's law of inertia
- Tensor products of vector spaces; universal property
- Exterior algebra; determinant as top exterior power; Pfaffian
- Symmetric powers; applications in physics, differential geometry, ML
- Tensor products as the foundation for all tensor decompositions (Part II)

1.9 9. Modules and Canonical Forms

- Modules over a ring; free modules; connection to vector spaces
- Structure theorem for finitely generated modules over a PID
- Jordan and rational canonical forms as corollaries
- Why this matters: unifies eigentheory; also underpins homological methods

2 PART II — LINEAR ALGEBRA (COMPUTATIONAL)

2.1 10. Floating Point and Numerical Stability

- IEEE arithmetic; machine epsilon; rounding models
- Condition number: of a matrix and of a problem
- Forward and backward error analysis; Wilkinson's approach
- Catastrophic cancellation; why naive algorithms fail

2.2 11. LU and Cholesky Factorization

- Gaussian elimination as LU factorization
- Partial and complete pivoting; stability analysis
- Cholesky for symmetric positive definite systems
- Solving $Ax = b$ stably; cost analysis
- Applications: scientific simulation, Bayesian linear systems

2.3 12. QR Factorization

- Modified Gram-Schmidt; why classical fails numerically
- Householder reflections; Givens rotations
- QR for least squares; normal equations vs. QR
- Rank-deficient least squares; pseudoinverse; regularization
- Applications: regression, overdetermined systems, data fitting

2.4 13. Singular Value Decomposition (Computational)

- Golub-Reinsch algorithm; bidiagonalization
- Truncated SVD; optimal low-rank approximation (Eckart-Young)
- Applications: PCA, image compression, latent semantic analysis, pseudoinverse
- Connection to randomized methods (Section 18)

2.5 14. Tensor Decompositions and Tensor Networks

- CP (CANDECOMP/PARAFAC) decomposition; alternating least squares
- Tucker decomposition; multilinear SVD (HOSVD)
- Tensor train (TT) / matrix product states; bond dimension

- NP-hardness of tensor rank; uniqueness conditions
- Applications: neural network compression, quantum many-body, high-dimensional PDEs

2.6 15. Eigenvalue Algorithms

- Power iteration; inverse iteration; shift-and-invert
- QR algorithm with shifts and deflation
- Symmetric QR with Wilkinson shift; divide-and-conquer
- Krylov subspace methods: Lanczos (symmetric), Arnoldi (general)
- Applications: Google PageRank, structural mechanics, Schrödinger operators

2.7 16. Iterative Methods for Linear Systems

- Krylov methods: CG (SPD), MINRES (symmetric), GMRES (general), BiCGSTAB
- Convergence theory; connection to polynomial approximation of eigenvalue distribution
- Preconditioning: ILU, incomplete Cholesky, algebraic multigrid (conceptual)
- CG as optimization; connection to Course 5

2.8 17. Sparse Matrices

- Storage formats: CSR, CSC, COO; arithmetic on sparse formats
- Sparse direct methods: fill-in, reordering (AMD, nested dissection)
- Sparse iterative methods in practice; choosing a solver
- Applications: large-scale PDE, graph problems, network analysis

2.9 18. Randomized Linear Algebra

- Randomized SVD (Halko-Martinsson-Tropp); single-pass variant
- Sketching and random projections; Johnson-Lindenstrauss lemma
- Randomized least squares; sketch-and-solve; sketch preconditioners
- Kaczmarz algorithm as SGD on normal equations
- Applications: large-scale ML, streaming data, massive linear systems

2.10 19. Spectral Graph Theory

- Graph Laplacian (combinatorial and normalized); spectrum examples
- Cheeger inequality: algebraic connectivity vs. graph expansion
- Random walks on graphs; stationary distributions; mixing times
- Spectral clustering; manifold learning (Laplacian eigenmaps, diffusion maps)
- PageRank; graph neural networks as spectral operators

2.11 20. Structured Matrices and Fast Algorithms

- Toeplitz, circulant, Hankel matrices; displacement rank
- FFT as diagonalization of circulant matrices
- Hierarchical matrices (H-matrices): structure and arithmetic (brief)
- Fast multipole method (conceptual); applications to N -body and integral equations

2.12 21. Matrix Functions and Matrix Calculus

- Matrix exponential: Padé approximation and scaling-squaring
- Other functions: square root, logarithm, sign; Krylov approximation
- Fréchet derivative of matrix functions
- Matrix calculus: gradients and Hessians of scalar matrix functions
- Applications: ODE solvers, network analysis, optimization on matrix manifolds

3 PART III — ABSTRACT ALGEBRA IN SERVICE OF APPLICATIONS

3.1 22. Groups and Symmetry

- Definitions and examples: symmetric, cyclic, dihedral, matrix groups
- Subgroups; Lagrange's theorem; cosets; normal subgroups
- Why this matters: symmetry is a modeling primitive in physics, chemistry, and ML

3.2 23. Group Homomorphisms and Quotients

- Homomorphisms and isomorphisms
- Quotient groups; isomorphism theorems

- Direct products; semidirect products
- Applications: symmetry breaking, gauge theory, signal processing on groups

3.3 24. Group Actions and Symmetry in Computation

- Actions, orbits, stabilizers; orbit-stabilizer theorem
- Burnside's lemma; counting with symmetry
- Cayley's theorem; conjugacy classes; class equation
- Applications: molecular symmetry, enumeration, equivariant architectures

3.4 25. Sylow Theory and Group Structure

- Sylow theorems: existence, conjugacy, number (statements and proofs)
- Classification of groups of small orders; direct applications
- Simple groups: A_5 is simple (one proof)
- Classification of finite abelian groups
- Applied payoff: understanding symmetry groups of physical systems; underpins representation theory

3.5 26. Rings and Arithmetic

- Definitions; examples ($\mathbb{Z}$, polynomial rings, matrix rings, $\mathbb{Z}/n\mathbb{Z}$)
- Ideals, quotient rings, ring homomorphisms; isomorphism theorems
- Product rings and Chinese Remainder Theorem (with algorithm)
- Prime and maximal ideals
- Applications: modular arithmetic, error-correcting codes, cryptography

3.6 27. Divisibility, Factorization, and Polynomial Rings

- $\text{ED} \rightarrow \text{PID} \rightarrow \text{UFD}$: the hierarchy; emphasis on payoffs
- Gaussian integers; applications to number-theoretic algorithms
- Polynomial rings: Eisenstein, reduction mod p
- Applications: irreducibility testing, polynomial system solving

3.7 28. Field Theory and Finite Fields

- Field extensions; degree $[K : F]$; algebraic elements and minimal polynomial
- Splitting fields; normal and separable extensions
- Finite fields $\text{GF}(p^n)$: structure, existence, primitive element
- Applications: coding theory (Reed-Solomon, LDPC), cryptography (AES, elliptic curves), pseudorandom generation

3.8 29. Galois Theory

- Galois group; fundamental theorem of Galois theory
- Computing Galois groups (examples over $\mathbb{Q}$)
- Solvable groups; solvability by radicals; quintic unsolvability
- Cyclotomic fields; roots of unity
- Applied payoff: underpins algebraic coding theory; explains why generic degree-5 polynomial systems require numerical rather than closed-form solution

3.9 30. Modules

- Modules over a ring; submodules; quotients; homomorphisms
- Free modules; exact sequences; Snake lemma
- Structure theorem for finitely generated modules over a PID
- Tensor product of modules; Hom functor; adjunction
- Applications: unifies linear algebra over arbitrary rings; foundation for homological algebra

3.10 31. Representation Theory of Finite Groups

- Group algebras $k[G]$; representations as G -modules
- Maschke's theorem: complete reducibility in characteristic 0
- Irreducible representations; Schur's lemma
- Characters and character tables; orthogonality relations
- Applications: Burnside's theorem ($p^a q^b$ solvable), Fourier analysis on finite groups, design of symmetric algorithms

3.11 32. Harmonic Analysis on Groups

- Pontryagin duality for locally compact abelian groups
- Peter-Weyl theorem for compact groups
- Spherical harmonics as irreducible representations of $SO(3)$
- Generalized Fourier analysis; convolution on groups
- Applications: fast algorithms on groups, non-commutative Fourier analysis, physics

3.12 33. Equivariant Maps and Geometric Deep Learning

- Equivariant and invariant linear maps; Schur's lemma computationally
- $SO(3)$ and $SE(3)$ representations; spherical harmonics explicitly
- Equivariant neural network architectures (E3NN, NequIP)
- Gauge equivariance in physics-inspired models
- Applications: molecular property prediction, protein structure, robotics

3.13 34. Gröbner Bases and Polynomial Systems

- Polynomial ideals; term orderings; Buchberger's algorithm
- Reduced Gröbner bases; ideal membership; elimination theory
- Solving polynomial systems; robotics kinematics; computer vision
- Software: SymPy, Macaulay2, Singular

3.14 35. Commutative Algebra and Algebraic Geometry (Brief)

- Noetherian rings; Hilbert basis theorem
- Localization; local rings; geometric meaning
- Prime spectrum $\text{Spec}(R)$; Hilbert's Nullstellensatz
- Applied payoff: foundations of algebraic statistics, sum-of-squares optimization, polynomial optimization

3.15 36. Homological Algebra and Topological Data Analysis

- Chain complexes and homology; long exact sequences
- Functors Hom and $\otimes$; derived functors Ext and Tor
- Simplicial complexes as concrete homological objects

- Persistent homology: filtrations, barcodes, persistence diagrams, stability theorem
- Software: Ripser, Gudhi, Giotto-TDA
- Applications: shape analysis, materials, neuroscience, time series topology